## APPENDIX IV. LESOTHO: EXTERNAL DEBT SUSTAINABILITY ANALYSIS[16]

*In the view of Fund and IDA staffs, Lesotho presents a moderate risk of debt distress. External debt in NPV terms is projected to remain well below 40 percent of GDP, both in the baseline scenario and under a set of adverse shocks. However, the DSA also highlights the country's vulnerability to exchange rate changes and other shocks, given the country's narrow export base.*

**Lesotho's nominal external debt declined from 82 percent of GDP in 2002/03 to 47 percent of GDP (US$694 million) at the end of 2005/06,** as a result of the combined effects of limited new borrowing, an appreciation of the exchange rate, and the early repayment of nonconcessional loans in 2005. The NPV of external debt stood at 31 percent of GDP at end-March 2006. About 85 percent of the debt is owed to multilateral creditors, mainly IDA (75 percent). The remainder is split between bilateral and commercial creditors (10 percent and 3 percent, respectively). Virtually all public and publicly-guaranteed external debt is held by the central government, which has taken over the debt incurred by other public entities. For private sector debt, only obligations towards countries outside the Common Monetary Area are recorded. At end-March 2006, these obligations were estimated at US$2.2 million.

**Macroeconomic projections and financing terms are summarized below in Box 1.** Compared to the 2005 DSA (see CR05/437), the current account balance is significantly improved, on account of larger than previously expected SACU revenues to 2009/10 and stronger export performance in the non-garment sector, primarily due to diamonds. Real GDP growth is projected to average about 2 percent to 2011 and to increase to 4.5 percent over the long term, driven by growth in non-traditional exports and reflecting the effects of the implementation of the PRS and other planned structural reforms.

**The baseline scenario assumes that the authorities would contain the fiscal deficit over the medium term to levels that can be financed with grants and highly concessional loans.** Gross domestic public debt currently stands at 6 percent of GDP, and is projected to decline further over the medium term. Moreover, net domestic debt is currently negative—at minus 10 percent of GDP, reflecting sizable government deposits.

**Under the baseline scenario, all debt burden indicators remain below their relevant indicative thresholds.**[17] Debt level indicators stabilize at low levels under the terms of new

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[16] Jointly updated with the World Bank staff. The Debt Sustainability Analysis (DSA) focuses on external debt. Domestic debt is not expected to significantly increase the total public debt burden in the long term.

[17] The latest World Bank Country Policy and Institutional Assessment (CPIA) ranks Lesotho as a "medium performer" in terms of policy and institutions. The applicable indicative thresholds for debt sustainability, proposed under the framework for low-income countries, are an NPV of debt of 40 percent of GDP, 150 percent

(continued)

borrowing assumed for the second half of the projection period. Reflecting in part the growth of nontextile exports including diamonds, the NPV of external debt in terms of exports declines steadily under the baseline scenario, with the ratio remaining well below the indicative threshold of 150 percent over the medium term.

**Under the standard stress tests, based on the volatility of key macroeconomic variables, the debt burden remains below the indicative thresholds.** In case of a large nominal devaluation (30 percent), the NPV of debt would exceed 40 percent of GDP, but only over the initial five years of the projection period (Table 1b, Scenario B6). In all other scenarios the level of debt consistently remains below the indicative thresholds. An additional test, designed to illustrate the vulnerabilities arising from the country's narrow export base,

**Box 1. Main Assumptions Under the Baseline Scenario**

- Annual Real GDP growth is assumed to increase gradually from about 2 percent in 2005/06 to 4.5 percent by the end of the projection period.
- Inflation (as measured by the GDP deflator) is projected to stabilize slightly above 4 percent, broadly in line with experience in recent years.
- Fiscal deficits are projected to remain contained at a level that can be financed through external donors' assistance. The government is assumed to pursue a fiscal policy that would help maintain low domestic debt relative to GDP
- Imports and exports of goods and services and transfers are assumed to grow in line with GDP (in U.S. dollar terms). FDI is assumed to grow slightly more rapidly, taking into account the reduction of the corporate income tax rate in 2006 to attract foreign investment.
- The current account deficit (including official transfers) is determined by the above trends, widening to about 5 percent of GDP in 2006/07 and broadly remaining at this level over the medium term.
- Net external public sector financing is assumed to rise to about 2 percent of GDP by the end of the current decade and then to stabilize at this level. Foreign grants are assumed to remain stable at about 3 percent of GDP over the projection period. The DSA assumes that new borrowing would be contracted on highly concessional terms during the first half of the projection period; however, as the country reaches the threshold for eligibility under concessional windows of financing, it is assumed that the average grant element of new borrowing gradually decreases from 45 percent to 20 percent by the end of the projection period.
- Private sector debt is projected to increase only marginally in terms of GDP, to 0.6 percent by 2024/25.

assumes that the expansion of textile exports since 1999 based on the United States' African Growth and Opportunity Act (AGOA) would be entirely reversed in 2006/07.[18] Under this extreme scenario, the debt sustainability indicators would rise sharply, with the NPV of debt-

of exports, and 250 percent of government revenues, and debt service of 20 percent of exports and 30 percent of government revenues.

[18] Test B7 in Table 1b.

to-exports ratio peaking at about 174 percent in 2007/08, but drop below their indicative thresholds over the medium term.

**Based on the findings of this DSA, staffs are of the view that Lesotho faces a moderate risk of debt distress.** Under the baseline scenario, all debt burden indicators are expected to remain below the relevant indicative thresholds. Stress tests, based on the country's historical volatility, do not indicate any breach of the thresholds. However, shocks affecting the country's narrow export base could increase the country's risk of debt distress.

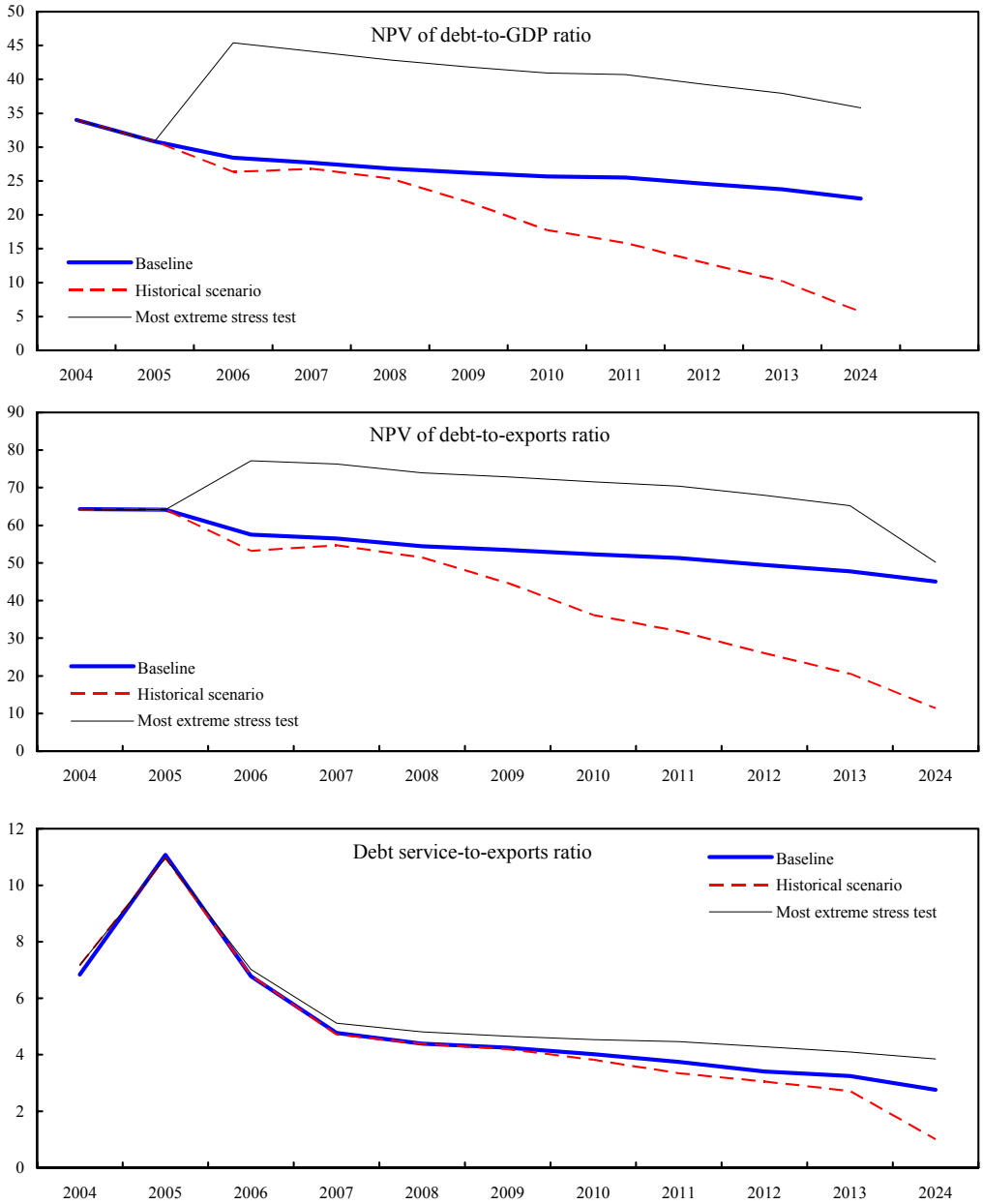
Table 1a. Lesotho External Debt Sustainability Framework, Baseline Scenario, 2001-2025 [1]
(In percent of GDP, unless otherwise indicated)

| | Actual | | | | | Historical Average [6] | Standard Deviation [6] | | | | | | 2006-11 Average | | | 2011-25 Average |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2001/02 | 2002/03 | 2003/04 | 2004/05 | 2005/06 | | | 2006/07 | 2007/08 | 2008/09 | 2009/10 | 2010/11 | | 2014/15 | 2024/25 | |
| **External debt (nominal)[1]** | **77.7** | **82.0** | **61.6** | **50.0** | **47.3** | | | **47.3** | **46.0** | **45.0** | **44.3** | **43.8** | | **39.4** | **31.7** | |
| *o/w* public and publicly guaranteed (PPG) | 77.4 | 81.8 | 61.4 | 49.8 | 47.2 | | | 47.1 | 45.8 | 44.8 | 44.1 | 43.6 | | 39.1 | 31.3 | |
| Change in external debt | 8.0 | 4.3 | -20.4 | -11.6 | -2.7 | | | 0.0 | -1.3 | -1.0 | -0.7 | -0.5 | | -1.2 | -0.5 | |
| Identified net debt-creating flows | 22.7 | 7.6 | -20.0 | -12.9 | -2.6 | | | 0.0 | -5.3 | -2.0 | 1.0 | 2.3 | | 0.0 | 0.1 | |
| **Non-interest current account deficit** | **12.2** | **16.5** | **9.7** | **2.0** | **1.0** | **17.2** | **10.4** | **3.2** | **-1.8** | **1.4** | **4.7** | **6.0** | | **4.5** | **4.5** | 4.5 |
| Excl. LHWP | -1.9 | -1.5 | -1.1 | -1.0 | -0.8 | -1.5 | 0.7 | -1.5 | -0.6 | -0.7 | -0.7 | -0.6 | | -0.4 | -0.5 | -0.4 |
| Deficit in balance of goods and services | 54.0 | 58.2 | 49.8 | 45.3 | 45.0 | | | 51.5 | 49.5 | 47.4 | 46.3 | 45.1 | | 43.2 | 43.2 | |
| Exports | 46.4 | 54.2 | 48.2 | 52.9 | 48.1 | | | 49.5 | 49.0 | 49.3 | 49.0 | 49.1 | | 49.7 | 49.7 | |
| Imports | 100.4 | 112.5 | 98.1 | 98.2 | 93.1 | | | 101.0 | 98.5 | 96.7 | 95.4 | 94.2 | | 93.0 | 93.0 | |
| Net current transfers (negative = inflow) | -17.2 | -17.2 | -15.6 | -19.5 | -21.4 | -18.0 | 2.1 | -24.7 | -29.2 | -24.1 | -20.4 | -18.8 | | -17.7 | -17.7 | -17.8 |
| Other current account flows (negative = net inflow) | -24.6 | -24.5 | -24.5 | -23.8 | -22.5 | | | -23.6 | -22.1 | -21.9 | -21.2 | -20.3 | | -21.0 | -21.0 | |
| **Net FDI (negative = inflow)** | **-3.7** | **-3.9** | **-3.9** | **-3.9** | **-3.9** | **-3.4** | **0.5** | **-3.6** | **-3.5** | **-3.4** | **-3.3** | **-3.2** | | **-3.2** | **-3.5** | -3.3 |
| **Endogenous debt dynamics[2]** | **14.2** | **-5.0** | **-25.7** | **-11.0** | **0.2** | | | **0.4** | **0.0** | **0.1** | **-0.5** | **-0.5** | | **-1.3** | **-0.8** | |
| Contribution from nominal interest rate | 1.9 | 1.5 | 1.1 | 1.0 | 0.8 | | | 1.5 | 0.6 | 0.7 | 0.7 | 0.6 | | 0.4 | 0.5 | |
| Contribution from real GDP growth | -2.7 | -2.6 | -1.8 | -1.3 | -0.6 | | | -1.1 | -0.6 | -0.6 | -1.1 | -1.1 | | -1.7 | -1.3 | |
| Contribution from price and exchange rate changes | 15.1 | -3.9 | -25.0 | -10.7 | -0.9 | | | … | … | … | … | … | | … | … | |
| **Residual (3-4) [3]** | **-14.7** | **-3.3** | **-0.5** | **1.3** | **0.0** | | | **0.0** | **4.0** | **1.0** | **-1.7** | **-2.8** | | **-1.3** | **-0.7** | |
| *o/w* exceptional financing | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| NPV of external debt[4] | … | 67.8 | 43.6 | 34.2 | 31.0 | | | 28.6 | 27.8 | 27.0 | 26.4 | 25.9 | | 23.5 | 22.8 | |
| In percent of exports | … | 125.0 | 90.4 | 64.6 | 64.5 | | | 57.8 | 56.8 | 54.8 | 53.8 | 52.7 | | 47.2 | 45.9 | |
| **NPV of PPG external debt** | **…** | **67.6** | **43.4** | **34.0** | **30.9** | | | **28.5** | **27.7** | **26.9** | **26.2** | **25.7** | | **23.2** | **22.4** | |
| In percent of exports | … | 124.7 | 90.1 | 64.3 | 64.2 | | | 57.5 | 56.5 | 54.5 | 53.5 | 52.3 | | 46.7 | 45.1 | |
| Debt service-to-exports ratio (in percent) | 12.3 | 8.0 | 7.5 | 7.1 | 10.3 | | | 7.0 | 4.9 | 4.6 | 4.4 | 4.2 | | 2.9 | 3.0 | |
| **PPG debt service-to-exports ratio (in percent)** | 12.2 | 7.9 | 7.4 | 6.8 | 11.1 | | | 6.8 | 4.8 | 4.4 | 4.2 | 4.0 | | 2.8 | 2.8 | |
| Total gross financing need (billions of U.S. dollars) | 101.0 | 131.4 | 107.2 | 26.1 | 30.9 | | | 83.9 | -37.8 | 4.9 | 66.1 | 92.2 | | 68.2 | 132.7 | |
| Non-interest current account deficit that stabilizes debt ratio | 4.2 | 12.2 | 30.1 | 13.6 | 3.7 | | | 3.2 | -0.5 | 2.3 | 5.5 | 6.5 | | 5.8 | 5.0 | |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 3.3 | 3.6 | 3.2 | 2.6 | 1.3 | 3.0 | 3.3 | 2.5 | 1.4 | 1.4 | 2.6 | 2.7 | 2.1 | 4.5 | 4.5 | 4.2 |
| GDP deflator in US dollar terms (change in percent) | -17.8 | 5.3 | 43.9 | 21.1 | 1.8 | 2.6 | 17.6 | 3.4 | 3.0 | 3.7 | 2.2 | 2.2 | 2.9 | 3.4 | 3.4 | 3.2 |
| Effective interest rate (percent)[5] | 2.3 | 2.0 | 2.0 | 2.1 | 1.6 | 2.2 | 0.6 | 3.4 | 1.4 | 1.6 | 1.6 | 1.5 | 1.9 | 1.1 | 1.7 | 1.2 |
| Growth of exports of G&S (US dollar terms, in percent) | 24.1 | 27.4 | 32.0 | 36.4 | -6.4 | 14.0 | 17.2 | 9.2 | 3.5 | 5.9 | 4.3 | 5.0 | 5.6 | 8.1 | 8.1 | 7.6 |
| Growth of imports of G&S (US dollar terms, in percent) | -6.5 | 22.2 | 29.4 | 24.4 | -2.3 | 4.8 | 16.4 | 15.0 | 1.9 | 3.3 | 3.4 | 3.6 | 5.5 | 8.1 | 8.1 | 7.4 |
| Grant element of new public sector borrowing (in percent) | … | … | … | 46.5 | 43.8 | … | … | 49.5 | 46.5 | 47.3 | 47.4 | 47.5 | 47.6 | 42.5 | 15.0 | 31.8 |
| *Memorandum item:* | | | | | | | | | | | | | | | | |
| Nominal GDP (millions of US dollars) | 709.6 | 773.9 | 1149.1 | 1427.8 | 1471.8 | | | 1560.2 | 1630.2 | 1715.5 | 1798.8 | 1887.4 | | 2275.5 | 5352.1 | |

Source: Staff simulations.

[1] Includes both public and private sector external debt.
[2] Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
[3] Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
[4] Assumes that NPV of private sector debt is equivalent to its face value.
[5] Current-year interest payments devided by previous period debt stock.
[6] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Figure 1. Lesotho: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2004-2024
(In percent)




Source: Staff projections and simulations.

Table 1b. Lesotho Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005-2025
(In percent)

| | | Projections | | | | | |
|---|---|---|---|---|---|---|---|
| | 2005/06 | 2006/07 | 2007/08 | 2008/09 | 2009/10 | 2014/15 | 2024/25 |
| **NPV of debt-to-GDP ratio** | | | | | | | |
| **Baseline** | 31 | 28 | 28 | 27 | 26 | **23** | 22 |
| **A. Alternative Scenarios** | | | | | | | |
| A1. Key variables at their historical averages in 2006-25 [1] | 31 | 26 | 27 | 25 | 22 | **8** | 6 |
| A2. New public sector loans on less favorable terms in 2006-25 [2] | 31 | 30 | 30 | 30 | 30 | **29** | 32 |
| **B. Bound Tests** | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2006-07 | 31 | 30 | 29 | 28 | 27 | **24** | 23 |
| B2. Export value growth at historical average minus one standard deviation in 2006-07 [3] | 31 | 28 | 28 | 27 | 26 | **23** | 22 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07 | 31 | 41 | 40 | 39 | 38 | **34** | 33 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 [4] | 31 | 38 | 37 | 36 | 36 | **31** | 25 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 31 | 42 | 41 | 40 | 39 | **34** | 29 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 [5] | 31 | 45 | 44 | 43 | 42 | **37** | 36 |
| B7. Loss of trade preferences (export ratio returning to 1999 level in 2006) [6] | 31 | 40 | 39 | 37 | 36 | **31** | 28 |
| **NPV of debt-to-exports ratio** | | | | | | | |
| **Baseline** | 64 | 57 | 56 | 54 | 53 | **47** | 45 |
| **A. Alternative Scenarios** | | | | | | | |
| A1. Key variables at their historical averages in 2006-25 [1] | 64 | 53 | 55 | 51 | 45 | **16** | 11 |
| A2. New public sector loans on less favorable terms in 2006-25 [2] | 64 | 60 | 61 | 60 | 61 | **59** | 64 |
| **B. Bound Tests** | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2006-07 | 64 | 57 | 56 | 54 | 53 | **47** | 45 |
| B2. Export value growth at historical average minus one standard deviation in 2006-07 [3] | 64 | 63 | 62 | 59 | 58 | **51** | 49 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07 | 64 | 57 | 56 | 54 | 53 | **47** | 45 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 [4] | 64 | 77 | 76 | 74 | 73 | **63** | 50 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 64 | 65 | 64 | 62 | 61 | **53** | 45 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2005 [5] | 64 | 57 | 56 | 54 | 53 | **47** | 45 |
| B7. Loss of trade preferences (export ratio returning in 2005 to 1999 level) [6] | 64 | 80 | 174 | 168 | 164 | **137** | 123 |
| **Debt service ratio** | | | | | | | |
| **Baseline** | 11 | 7 | 5 | 4 | 4 | **3** | 3 |
| **A. Alternative Scenarios** | | | | | | | |
| A1. Key variables at their historical averages in 2006-25 [1] | 11 | 7 | 5 | 4 | 4 | **2** | **1** |
| A2. New public sector loans on less favorable terms in 2006-25 [2] | 11 | 7 | 5 | 5 | 5 | **4** | 4 |
| **B. Bound Tests** | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2006-07 | 11 | 7 | 5 | 5 | 4 | **3** | 3 |
| B2. Export value growth at historical average minus one standard deviation in 2006-07 [3] | 11 | 8 | 5 | 5 | 5 | **3** | 3 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07 | 11 | 7 | 5 | 5 | 4 | **3** | 3 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 [4] | 11 | 7 | 5 | 5 | 5 | **4** | 3 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 11 | 6 | 5 | 4 | 4 | **3** | 3 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 [5] | 11 | 7 | 5 | 5 | 4 | **3** | 3 |
| B7. Loss of trade preferences (export ratio returning to 1999 level in 2006) [6] | 11 | 34 | 22 | 15 | 14 | **10** | 8 |
| *Memorandum item:* | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) [7] | 38 | 38 | 38 | 38 | 38 | **38** | 38 |

Source: Staff projections and simulations.

[1] Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
[2] Assumes that the interest rate on new borrowing is 2 percentage points higher than in the baseline, while grace and maturity periods are the same.
[3] Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assumming an offsetting adjustment in import levels).
[4] Includes official and private transfers and FDI.
[5] Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
[6] This test was added for Lesotho, and is not part of the standard set of bound tests.
[7] Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

## APPENDIX V. LESOTHO: SOCIAL AND DEMOGRAPHIC INDICATORS

| **Population** | |
|---|---|
| **Total Population** | 1.8 million |
| Population growth rate | 0.9 |
| Life expectancy at birth (in years) | 37.2 |
| Infant mortality rate (per thousand) | 79 |
| Urban Population (% total) | 18.1 |
| Population younger than 15 (Percent of total population) | 41.1 |
| **GDP per capita** (in US dollars) | 600 |
| **Access to safe water (2002)** | |
| Percent of total population | 76 |
| Urban | 88 |
| Rural | 74 |
| **Labor Statistics** (in thousands) | |
| Labor force (thousands) | 756 |
| Public sector employment (thousands, FY 2004/05) | 35.9 |
| Unemployment rate (percent) | 30 |

| **Geographical features** | |
|---|---|
| Area (sq. km.) | 30,350 |
| Population density (per sq. km.) | 59.6 |
| Arable land (2002, percent of total) | 10.9 |
| **Health** | |
| Population per physician (2000) | 16,436 |
| Public health expenditure (2002, in percent of GDP) | 5.3 |
| HIV prevalence rate (ages 15-49) | 28.9 |
| **Education (2002)** | |
| Adult literacy rate (percent) | 81.4 |
| Primary school enrollment (net, percent) | 85.8 |
| **Poverty Indicators (1995)** | |
| Share of income, lowest 20 percent (percent) | 1.5 |
| GINI Index | 63.2 |

Sources: World Bank, 2005, *World Development Indicators 2005*, and national authorities. Data refer to 2003, unless otherwise indicated.

**Statement by the IMF Staff Representative**
**October 6, 2006**

This statement provides an update on economic developments since the issuance of the staff report. These developments do not change the thrust of the staff appraisal.

- The Southern African Customs Union (SACU) Council has indicated its intention to distribute additional windfall SACU revenue during fiscal year 2006/07. The magnitude and timing of the receipts are expected to be determined in October 2006. The authorities stated that they plan to pursue a prudent fiscal policy and resist pressures to spend.

- The exchange rate of the loti against the US dollar moved from M7.0 at the end of July 2006 to M7.6 at end-September 2006, in line with the weakening of the South African rand/US dollar exchange rate.

- Gross international reserves rose from US$564 million, representing about 5 months coverage of import of goods and nonfactor services at end-March, to US$604 million at end-September 2006, representing 5.2 months of coverage.